Technical Progress Tracking and ATS Pipeline Development for SkillSync: An AI-Powered Resume Screening and Career Matching System

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Abstract— This progress report documents the implementation status of SkillSync, an AI-powered resume screening and career matching system. The project follows a structured phase-based development cycle with regular progress updates to the assigned guide. The completed work covers workspace and architecture setup, React/Vite frontend development, Node.js/Express backend development, MongoDB integration, JWT-based authentication and role handling, the student module, resume submission, text-based resume processing, technical skill detection, ATS-oriented scoring, section evaluation, missing-skill identification, and improvement recommendations. The current system provides an end-to-end student resume-analysis workflow. This report presents the implemented technical deliverables, current working architecture, sprint-wise progress, issues mitigated, and the next planned development stage.

Keywords— SkillSync, Resume Screening, ATS, Resume Analysis, Skill Detection, JWT, React, Node.js, Express, MongoDB, Python Service, Project Progress

I. INTRODUCTION

SkillSync is being developed as a web-based platform for automated resume screening and career assistance. The project aims to reduce manual effort in resume evaluation and provide students with understandable feedback about their resume quality, detected technical skills, missing skills, and areas for improvement. The implementation is being carried out incrementally so that each technical stage can be demonstrated and reviewed before proceeding to the next stage.

The current development cycle has reached the resume-analysis stage. The core application architecture, authentication, student workflow, and ATS-oriented analysis pipeline are operational in the local development environment. The remaining phases will extend the existing candidate profile toward job matching, skill-gap analysis, course recommendations, recruiter functionality, administration, testing, and final preparation.

II. CURRENT IMPLEMENTATION ARCHITECTURE

The implemented architecture consists of a React/Vite frontend, a Node.js/Express backend, MongoDB as the persistent data layer, and a separate Python analysis/ML service. The frontend communicates with the backend through APIs. The backend manages application logic, authentication and database interaction, while the Python service provides an independent layer for resume-analysis functionality.

Layer	Technology	Current Technical Responsibility
Frontend	React + Vite	User interface, authentication screens, student dashboard, resume upload and ATS-result display
Backend	Node.js + Express	REST APIs, authentication, authorization, application logic and service communication
Database	MongoDB + Mongoose	Application and user-related persistent data
Analysis Service	Python	Resume text processing and analysis-related functionality

Implemented service endpoints: Backend health check: GET /api/health. Python analysis service health check: GET /health. Current local services are frontend on port 5173, backend on port 5000, Python service on port 8000, with MongoDB connected to the SkillSync database.

III. IMPLEMENTATION & OPTIMIZATION

A. Project Foundation

- Created the main SkillSync workspace and separated frontend, backend, database and analysis-service responsibilities.

- Configured the React/Vite frontend, Node.js/Express backend and Python analysis service.
- Connected the application with MongoDB and verified the development environment.
- Established service health checks for backend and Python analysis service.

B. Authentication and Role Handling

- Implemented registration and login workflow.
- Implemented JWT-based authentication.
- Implemented protected-access handling and role-aware application structure.
- Integrated authentication between frontend and backend.

C. Student Module

- Implemented the student dashboard and student-side navigation.
- Implemented resume upload functionality.
- Connected resume submission from the frontend to backend APIs.
- Connected the uploaded resume with the processing pipeline and displayed the resulting analysis.

D. Resume Processing and ATS Pipeline

- Implemented the current text-based and regular-expression based resume-processing approach.
- Implemented technical skill detection and keyword-oriented evaluation.
- Implemented overall ATS-style scoring and component-level scoring.
- Implemented section completeness, experience, education and project evaluation.
- Implemented missing-skill identification and improvement recommendations.
- Implemented structured ATS-result presentation in the frontend.

IV. EXPERIMENTAL / CURRENT RESULTS

The current implementation provides a working end-to-end resume-analysis workflow. After authentication, a student can upload a resume and receive a structured ATS dashboard. The current output contains overall ATS score, keyword score, skills score, section completeness, experience evaluation, education evaluation, project evaluation, detected technical skills, missing skills and improvement recommendations.

In the demonstrated current result, the system successfully detected 21 technical skills from the uploaded resume. The result is presented as an explainable analysis rather than only a single score, allowing the user to understand the individual factors contributing to the resume evaluation.

Current Feature	Implementation Status
Resume upload	Completed
Resume text processing	Completed
Technical skill detection	Completed
Keyword analysis	Completed
ATS score	Completed
Skills score	Completed
Section completeness	Completed
Experience / Education / Projects evaluation	Completed
Missing skills	Completed
Improvement recommendations	Completed

V. DEVELOPMENT PROGRESS UPDATE

The following sprint-style status report records the completed technical deliverables in the same formal progress-tracking style used for the project reference document. It is intended to be updated during regular meetings with the assigned guide.

Sprint	Focus	Technical Deliverables	Status
Sprint 1	System Foundation	Workspace setup; React/Vite	Completed

		frontend; Express backend; Python service; MongoDB connection; service verification	
Sprint 2	Authentication	Registration/login; JWT authentication; protected access; role handling	Completed
Sprint 3	Student Module	Student dashboard; resume upload; frontend-backend integration; result display	Completed
Sprint 4	Resume + ATS	Resume processing; skill detection; keyword analysis; ATS scoring; section/experience/education/project evaluation; missing skills; recommendations	Completed
Sprint 5	Job Matching	Job requirements, candidate-job matching and skill-gap foundation	Pending
Sprint 6	Recruiter Module	Recruiter dashboard and candidate workflow	Pending
Sprint 7	Admin Module	Administration and management functions	Pending
Sprint 8	Testing + Polish	System testing, regression fixes and UI polishing	Pending

A. Current Roadblocks Successfully Mitigated

- Frontend runtime issue encountered during development was resolved.
- Backend port/service startup conflicts were resolved.
- MongoDB connection and integration issues were resolved.
- Service health and communication were re-verified after fixes.

B. Current Project Completion

According to the original implementation plan, Phases 0 through 4 are completed. This corresponds to approximately 4 of the 9 major implementation stages and places the project at roughly the 45–50% overall implementation point. The completed portion includes the core technical foundation and a working resume-screening pipeline.

VI. NEXT DEVELOPMENT STAGE

The immediate next stage is Phase 5: Job Matching + Skill Gaps + Courses. The existing detected skills and resume profile will be used as the foundation for comparing candidates with job requirements. The planned work includes job-description handling, candidate-job matching, skill-gap identification, and the foundation for personalized course recommendations.

VII. CONCLUSION AND FUTURE SCOPE

The current development of SkillSync successfully establishes the technical foundation and a functioning resume-screening workflow. The implemented system can authenticate users, support the student workflow, accept resumes, process textual content, detect technical skills, generate ATS-oriented evaluation, identify missing skills, and provide improvement recommendations. The architecture is modular and prepared for the next stages of development.

Future work will focus on job matching, skill-gap analysis, course recommendations, recruiter functionality, administration, comprehensive testing, UI refinement, seed data, documentation and final demo preparation.